

PRACTICAL 14:

Aim: Design MVC Application to perform CRUD operation.

Step 1: Create MVC Project

1. Open Visual Studio
2. File → New → Project
3. Select:

ASP.NET Web Application (.NET Framework)

4. Click OK
5. Select: Empty
 - ✓ MVC checkbox
6. Click Create

Create a new ASP.NET Web Application

**Empty**
An empty project template for creating ASP.NET applications. This template does not have any content in it.

**Web Forms**
A project template for creating ASP.NET Web Forms applications. ASP.NET Web Forms lets you build dynamic websites using a familiar drag-and-drop, event-driven model. A design surface and hundreds of controls and components let you rapidly build sophisticated, powerful UI-driven sites with data access.

**MVC**
A project template for creating ASP.NET MVC applications. ASP.NET MVC allows you to build applications using the Model-View-Controller architecture. ASP.NET MVC includes many features that enable fast, test-driven development for creating applications that use the latest standards.

**Web API**
A project template for creating RESTful HTTP services that can reach a broad range of clients including browsers and mobile devices.

**Single Page Application**
A project template for creating rich client side JavaScript driven HTML5 applications using ASP.NET Web API. Single Page Applications provide a rich user experience which includes client-side interactions using HTML5, CSS3, and JavaScript.

Authentication
No Authentication
[Change](#)

Add folders & core references
☐ Web Forms
☒ MVC
☐ Web API

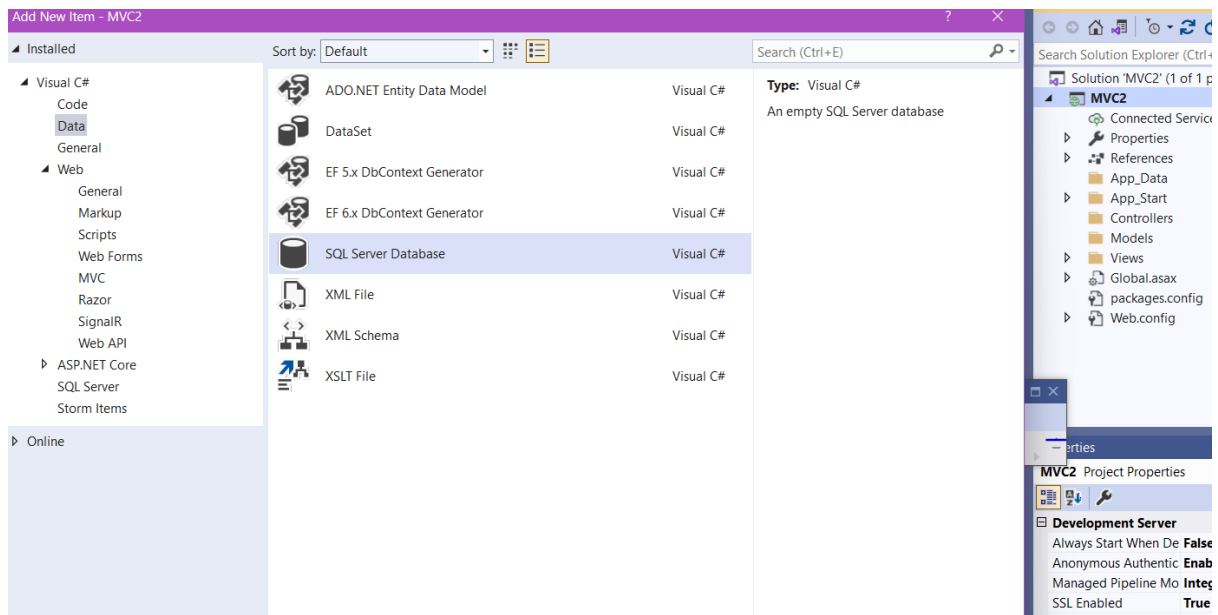
Advanced
☒ Configure for HTTPS
☐ Docker support
(Requires [Docker Desktop](#))
☐ Also create a project for unit tests

MVC2.Tests

Back

Create

Step 2: Create Database Table



Run SQL Query:

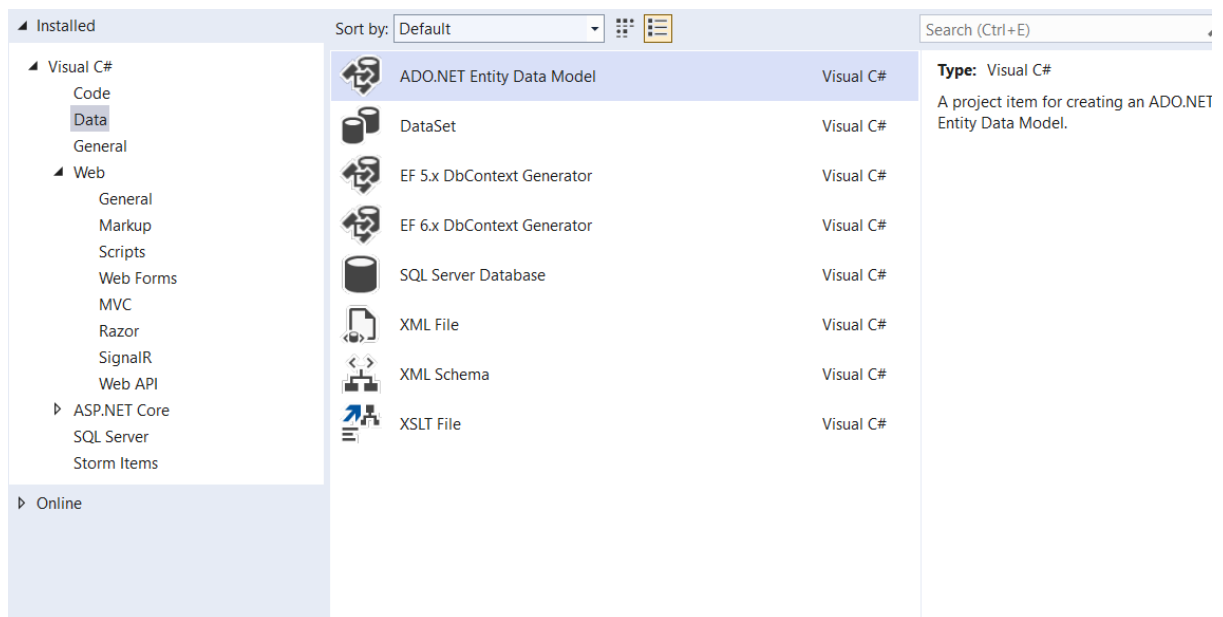
```
CREATE TABLE product (
    pid INT PRIMARY KEY,
    pname VARCHAR(100),
    pcost VARCHAR(100)
);
```

Insert records:

```
insert into product values(1,'Laptop','50000')
insert into product values(2,'Mouse','500')
```

Step 3: Add Entity Data Model

1. Right click Project
2. Add → New Item
3. Select: ADO.NET Entity Data Model
4. Click Add



Step 4: Configure Entity Framework

1. Select: EF Designer from Database
2. Click Next
3. Create database connection
4. Select database
5. Click Next
6. Select table: product
7. Click Finish



Choose Your Data Connection

Which data connection should your application use to connect to the database?

Database1.mdf

New Connection...

This connection string appears to contain sensitive data (for example, a password) that is required to connect to the database. Storing sensitive data in the connection string can be a security risk. Do you want to include this sensitive data in the connection string?

- ☐ No, exclude sensitive data from the connection string. I will set it in my application code.
- ☐ Yes, include the sensitive data in the connection string.

Connection string:

```
metadata=res://*/Model1.csdl|res://*/Model1.ssdl|
res://*/Model1.msl;provider=System.Data.SqlClient;provider connection string="data source=
(LocalDB)\MSSQLLocalDB;attachdbfilename=C:\Users\amnas\source\repos\MVC1\App_Data
\Database1.mdf;integrated security=True;connect
timeout=30;MultipleActiveResultSets=True;App=EntityFramework"
```

☒ Save connection settings in Web.Config as:

Database1Entities

< Previous

Next >

Finish

Cancel

☒ Tables
☐ Views
☐ Stored Procedures and Functions

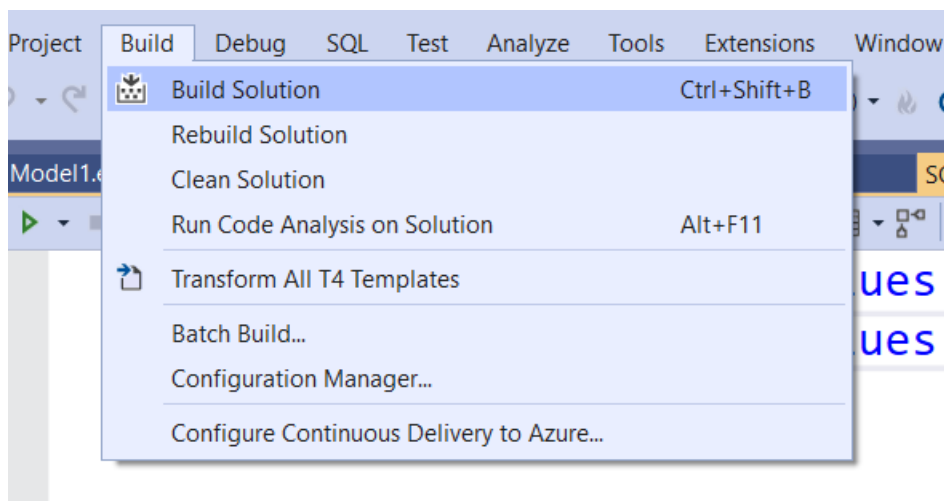
☒ Pluralize or singularize generated object names
☒ Include foreign key columns in the model
☐ Import selected stored procedures and functions into the entity model

Model Namespace:
Database1Model

< Previous Next > Finish Cancel

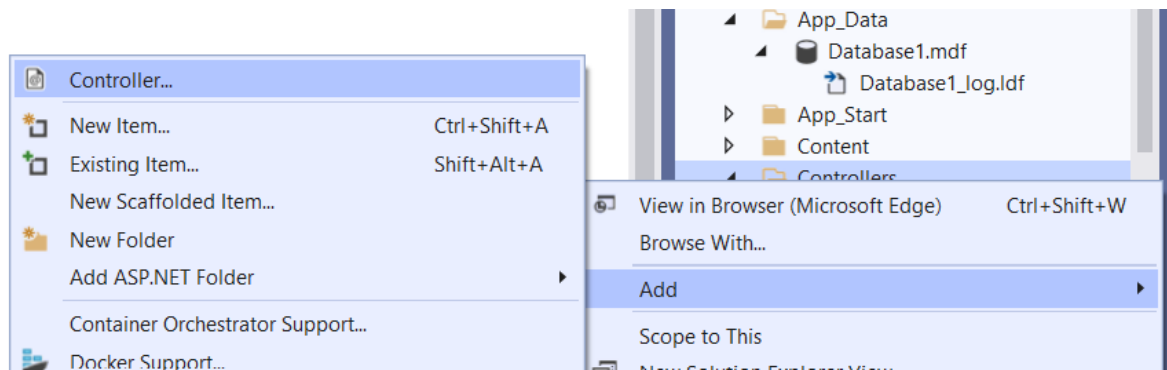
Step 5: Build Solution

1. Go to: Build → Build Solution



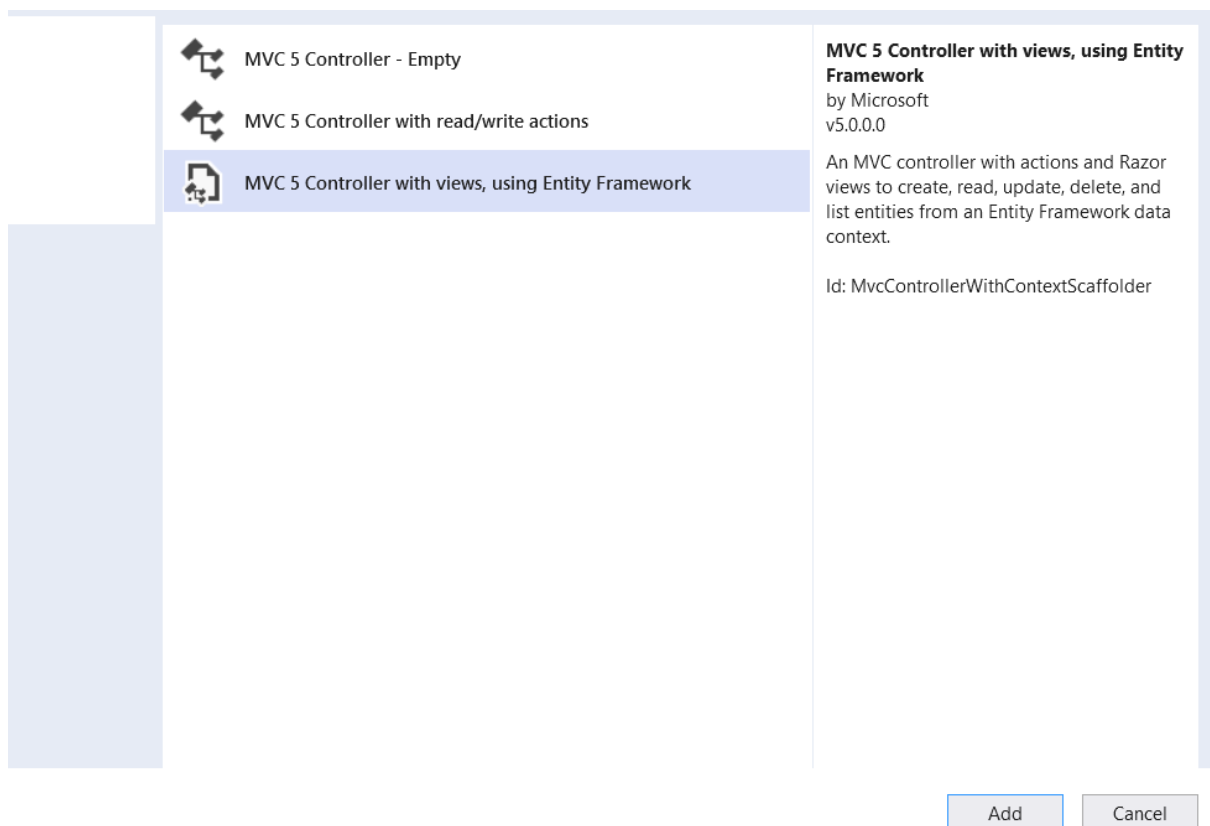
Step 6: Add Controller

1. Right click Controllers
2. Add → Controller



3. Select: MVC 5 Controller with views, using Entity Framework

4. Click Add



Step 7: Configure Controller

1. Model class: product
2. Data Context: Database1Entities
3. Controller Name: productsController
4. Click Add

Add Controller

Model class:

product (MVC1)

Data context class:

Database1Entities (MVC1)

+

☐ Use async controller actions

Views:

☒ Generate views

☒ Reference script libraries

☒ Use a layout page:

...

(Leave empty if it is set in a Razor _viewstart file)

Controller name:

productsController

Add

Cancel

Step 8: Run Project

1. Press F5
2. Open:

<https://localhost:xxxx/products>

Output:

←

↺

🔒 <https://localhost:44324/products/Index>

Sum

Application name

Home

About

Contact

Index

Create New

pname	pcost	
Laptop	50000	Edit Details Delete
Mouse	500	Edit Details Delete

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